

phys. stat. sol. (a) **91**, 205 (1985)

Subject classification: 14.4; 14.3.4; 22.9

Department of Postgraduate Studies and Research in Physics, Rani Durgavati University, Jabalpur¹⁾

Thermally Stimulated Depolarization Currents in Polypropylene Foils

By

J. M. KELLER and S. C. DATT

A careful study is undertaken of contactless thermally stimulated depolarization currents (TSDC) in unilaterally metallized commercial polypropylene foils, 15 μm thick. The films are charged both, positively and negatively by Townsend breakdown at fields ranging from 2 to 10 kV/cm. The TSDC thermograms are characterized by only one peak located at 393 K. The total charge released under the peak and peak current are found to increase nonlinearly with the polarizing field. Further, the peak current and total charge released under the peak are found to saturate at higher forming fields. These observed results are analysed in terms of immobilisation of space charges in the polymer. The samples charged at negative forming fields are found to decay slowly indicating that negative carriers are more stably trapped in polypropylene and hence to form stable electrets, polypropylene should be negatively charged.

Kontaktlose thermisch stimulierte Depolarisationsströme (TSDC) in lateral metallisierten kommerziellen, 15 μm -dicken Polypropylenschichten werden sorgfältig untersucht. Die Schichten werden sowohl positiv als auch negativ mittels Townsend-Durchbruchsfeldern zwischen 2 bis 10 kV/cm geladen. Die TSDC-Thermogramme sind durch nur ein Maximum bei 393 K charakterisiert. Es wird gefunden, daß die befreite Gesamtladung unter dem Maximum und der Maximumstrom nichtlinear mit dem Polarisationsfeld anwachsen. Es wird weiterhin gefunden, daß Maximumstrom und befreite Gesamtladung unter dem Maximum sich bei höheren Formierungsfeldern sättigen. Diese beobachteten Ergebnisse werden mit der Festsetzung von Raumladungen im Polymer diskutiert. Es wird gefunden, daß die bei negativen Formierungsfeldern geladenen Proben langsam abklingen, das zeigt, daß die negativen Ladungsträger im Polypropylen stabiler eingefangen sind und Polypropylen negativ geladen werden sollte, um stabile Elektrete zu erhalten.

1. Introduction

The interest in carrier storage and decay mechanisms in polymers arises not only from purely scientific reasons but also because of their practical importance in electrophotography [1] and electrostatic recording [2]. Recently, thermally stimulated discharge (TSD) of electrets has been widely used to investigate the nature of these mechanisms in polymeric dielectrics. The present paper reports the results of contactless thermally stimulated depolarization current measurements in polypropylene foils charged by the Townsend breakdown technique.

2. Experimental

Circular specimens of polypropylene, 3.6 cm in diameter and 15 μm in thickness, cut from commercial films supplied by Du Pont (USA) were used for the present investigation. Silver electrodes were evaporated on one side of the samples to provide as intimate contact as possible with the dielectric material.

¹⁾ Jabalpur (MP) 482001, India.

The contactless TSDC cycle involving an air gap between the sample and the upper electrode was obtained in the following manner. The sample was placed inside an oven which was fitted with "Laxson's Programmable Temperature Controller, type 1600-2/B".²⁾ With this instrument it was possible to achieve a linear heating rate of 3K/min. First, the oven and with it the sample was allowed to achieve an equilibrium temperature equal to the forming temperature which was chosen to be 413 K in each case. The non-metallized side of the specimen was Townsend charged by introducing a 50 μm thick air gap between the non-metallized face of the sample and the non-adhering metal electrode and then applying a field E_p (2 to 10 kV/cm). The time of polarization was adjusted to be 5 min in each case. The sample was then cooled to room temperature with the field still applied. Afterwards, the field was switched off and the sample was short-circuited for an arbitrary time of 5 min so as to remove the frictional and stray charges present. Thereafter, the sample was heated at a uniform rate of 3 K/min from room temperature to about 425 K and the current was recorded. The temperature of the sample was recorded with a precalibrated chromel-alumel thermocouple attached in close neighbourhood of the sample.

3. Theory of Thermally Stimulated Depolarization Current

The polarization in a dielectric is caused by (i) orientation of dipoles, (ii) transference of space charge, and (iii) injected surface charge from electrodes. In natural course, this polarisation decays very slowly and may take typically as long as many years for complete destruction of polarisation. Pillai and Goel [3] have attributed it to the trap hopping and relaxation of dipoles. When the electret is heated at a uniform rate in the absence of any field, the dipolar relaxation becomes smaller and smaller and a current is obtained which is called thermally stimulated depolarization current. The current increases progressively, reaches a maximum as the dipoles are gradually removed from their preferred orientation and then decreases. Similarly, the trapped space charges and injected surface charges give characteristic peaks at temperatures at which the thermal energy is sufficient to exhaust all the trapped charges at that level.

According to Bucci and Fieschi [4] the depolarization current at any time t due to dipolar reorientation on heating a polarized dielectric specimen is given by

$$j(t) = A \exp \left[\frac{-W}{kT} - r\tau_0^{-1} \int_{T_0}^T \exp \left(\frac{-W}{kT} \right) dt \right], \quad (1)$$

where A is a constant, W is activation energy, k is the Boltzmann constant, T_0 is the room temperature, and T is the temperature attained by the specimen in time t on heating it at a definite linear rate r , and τ_0 is a characteristic relaxation time given by the expression

$$\tau = \tau_0 \exp \left(\frac{W}{kT} \right), \quad (2)$$

where τ is the relaxation time of dipoles at the temperature T . The relation represented by (1) shows a peak at T_m which has the following relation with τ_0 :

$$\tau_0 = \frac{kT_m^2}{rW \exp(W/kT_m)}. \quad (3)$$

The above expression has been obtained by differentiating (1) and equating dj/dt to zero after substituting the value of the constant A . It is evident from (3) that T_m , the

²⁾ Supplied by Laxson's Electronic Pvt. Ltd., Bombay, India.

peak temperature is independent of both the polarizing field E_p and the polarizing temperature T_p , but depends upon the rate of heating r .

Expressions identical to (1) but with different values of the pre-exponential constant A have been given by Perlman [5] for discharge currents due to transference of space charges and trapped ions.

The total charge released during TSD is given by

$$Q = \int_0^{\infty} j(t) dt \quad (4)$$

and can be calculated by the area under the TSDC curve.

The activation energy W can be determined from the low temperature tail of (1) using

$$\log j = \log A - \frac{W}{kT}. \quad (5)$$

Thus a plot of $\log j$ versus $1000/T$ gives a straight line whose slope gives value of W . This is called initial rise method of Garlick and Gibson [6].

4. Results and Discussion

In general, below T_g semicrystalline polymers like polypropylene (PP) show two TSDC peaks in the amorphous region restrained by the crystallite and amorphous regions free from stress caused by the presence of crystallites [7]. However, above T_g experimental evidence of occurrence of other phase transitions is also reported in the literature [8]. But in the present investigation TSDC thermograms are characterized by only one peak located at 393 K as shown in Fig. 1. The total charge released in the external circuit was determined by calculating the time integral $\int j(t) dt$ over the depolarization current by Simpson's rule. The total charge released under the peak has been found to increase nonlinearly with the polarizing field. A typical curve between the total charge released and the applied field is shown in Fig. 2. However, the peak current and hence the charge stored under the peak have been observed to reach saturation at higher forming fields as shown in Fig. 3. These characteristics clearly indicate that the observed peak can be considered to be due to immobilization of space charges.

Such space charges are considered to be created because of the trapping of charge carriers showering over the non-metallized face of the sample and getting penetrated into the sample. The charge carriers may be getting trapped in surface and bulk traps of various depths. The space charge peak usually known as q peak in the literature is

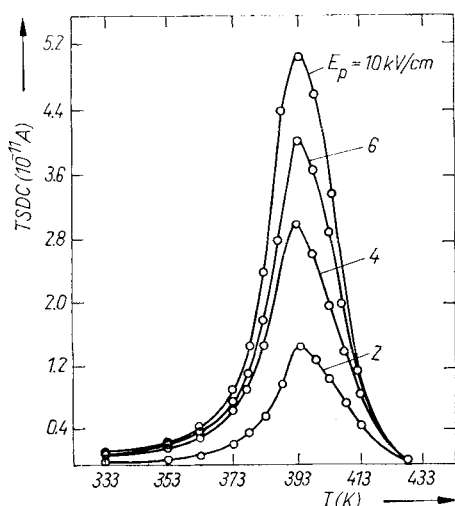


Fig. 1. Thermally stimulated discharge current (TSDC) thermograms for films charged at 413 K, with different poling fields

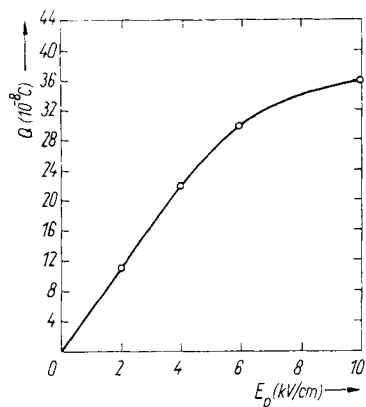


Fig. 2

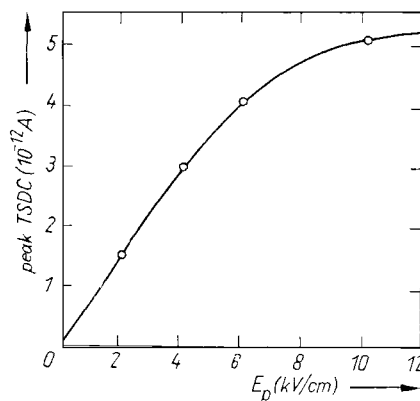


Fig. 3

Fig. 2. Variation of total charge Q released during TSDC cycle with poling fields, E_p , for films charged at 413 K

Fig. 3. Variation of peak TSDC with poling fields for films charged at 413 K

reported to depend nonlinearly on the polarizing field [9], as has been observed in the present case.

Usually a space-charge peak decreases in height at sufficiently high poling fields. However, in the present case, the peak height after attaining saturation does not show these characteristics. This decrease in height at higher forming fields is interpreted in terms of the self-motion of space charges accompanied by neutralization by recombination with thermally generated carriers. The carriers are generated uniformly in the entire specimen by dissociation of neutral entites.

As shown by Sessler [10] the discharge current is given by

$$j(t) = \frac{d[\sigma(t) - P(t)]/dt}{1 + d_1/d}, \quad (6)$$

where $\sigma(t) - P(t)$ is the effective charge density of the polymer, d_1 is the air gap thickness, and d is the sample thickness. Thus, the discharge current consists of a dipolar part $dP(t)/dt$ and a conduction part $d\sigma(t)/dt$.

Polypropylene is a nonpolar polymer, however, the possibility of impurity dipoles and so the contribution of the dipolar part can not be excluded. During charging the orientation of such dipoles leads to weakening of the field responsible for ohmic conduction across the sample. Thus, an increasing value of $dP(t)/dt$ is expected to limit the value of $d\sigma(t)/dt$ at high forming fields. This leads to the same charge release and hence the same peak height for electrets prepared at high temperature with large values of poling fields.

The observed peak in the present case, which corresponds to homocharging by air breakdown is quite sharp and well defined. These features can be considered to indicate the trapping of charge carriers in trapping centres distributed over a narrow energy range. This is further supported by the observed activation energy calculated by the initial rise method (5), the change in which is typical of a distribution of traps. Corresponding plots between $\log j$ and $1000/T$ is shown in Fig. 4. Presently, the initial slope of the peak increases slightly giving rise to an activation energy W which varies from 1.0 to 1.2 eV. Thus, the present study indicates the existence of trapping centres corresponding to trap depth ranging from 1.0 to 1.2 eV.

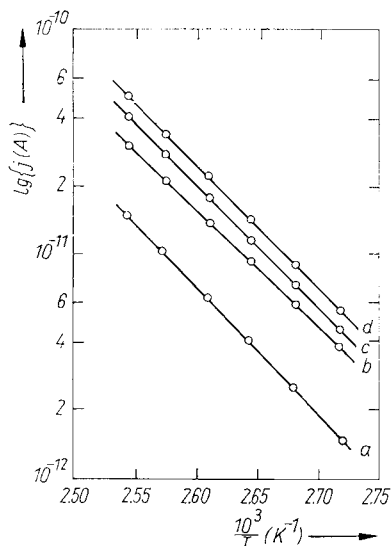


Fig. 4

Fig. 4. $\log j$ versus $10^3/T$ plots for films charged at 413 K, with different poling fields. (a) $E_p = 2$, (b) 4, (c) 6, (d) 10 kV/cm

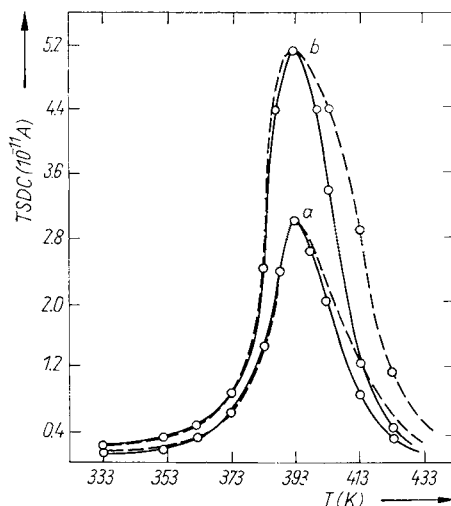


Fig. 5

Fig. 5. TSDC thermograms for films charged at 413 K, with different poling fields of different polarity. (a) 10 kV/cm (—), -10 kV/cm (---); (b) 4 kV/cm (—), -4 kV/cm (---)

Fig. 5 shows the TSDC thermograms of specimens which were Townsend charged with fields of different polarity. It is evident from these plots that the charge decay of electrets charged negatively is lower than that of electrets positively charged. This observed field dependence indicates that negative carriers are more stably trapped in this polymer and hence to form stable electrets, polypropylene should be negatively charged. In case of teflon FEP, such a field dependence has been reported by Turnhout [9] and Chudleigh et al. [11]. This strong charge stability for negative carriers may be due to the electronegativity of the methyl group present in this polymer and also due to the symmetry along the chains. Further, electrons can also be trapped in highly ordered regions of this semicrystalline polymer and also at the crystallite-amorphous interface [12]. It has already been reported in the literature that unsaturation sites along main chains act as traps in the bulk. Although, an exponential distribution in energy of traps has figured prominently in the literature, the presence of discrete traps in polymers has also been indicated by TSDC studies with negligible background [13]. Carrier transport in the bulk takes place by trap hopping with effective mobility governed by shallow traps. The positive carriers are considered to penetrate less, and hence these are supposed to be trapped in surface states only [14]. Due to the trapping of positive carriers in a near surface region, these are supposed to be easily neutralized by surface recombination.

5. Conclusion

On the basis of the observed characteristics of the TSDC peak which is located at 393 K, it can be concluded that this peak corresponds to immobilisation of space charges inside the polypropylene. Further, since the electrets charged at negative fields decay slowly as compared to those positively charged, to form stable electrets, polypropylene should be negatively charged.

Acknowledgements

The authors are grateful to Prof. T. S. Murty, formerly head of the department and now director, Madhya Pradesh Council of Science and Technology, Bhopal for encouragement. The authors are thankful to Dr. Ranjit Singh for various discussions. One of the authors (J.M.K.) is grateful to C.S.I.R. for the award of a post-doctoral fellowship.

References

- [1] M. CASSIERS, *Phot. Sci. Engng.* **4**, 199 (1960); *J. Phot. Sci.* **10**, 57 (1962).
- [2] U. ROTHGORDT, *Philips tech. Rdsch.* **36**, 98 (1976/77).
- [3] P. K. C. PILLAI and M. GOEL, *J. appl. Phys.* **44**, 3821 (1973).
- [4] C. BUCCI and R. FIESCHI, *Phys. Rev. Letters* **12**, 16 (1964).
- [5] M. M. PERLMAN, *Electrets*, The Electrochem Soc. 1973.
- [6] G. F. J. GARLICK and A. F. GIBSON, *Proc. Phys. Soc.* **60**, 574 (1948).
- [7] R. F. BOYER, *Encyclopedia of Polymer Science and Technology*, Suppl., Vol. II, Boyer and Boyer, Midland 1977 (p. 745).
- [8] R. F. BOYER, *J. Macromol. Sci. Phys. (B)* **3-4**, 503 (1973).
- [9] J. VAN TURNHOUT, *Thermally Stimulated Discharge of Polymer Electrets*, Elsevier Publ. Co., Amsterdam 1975.
- [10] G. M. SESSLER, *Electrets*, Springer-Verlag, New York 1980 (p. 167).
- [11] P. W. CHULDLEIGH, *Appl. Phys. Letters* **23**, 211 (1973).
- [12] R. A. CRESWELL, *Telesis (Canada)* **2**, 21 (1971).
- [13] M. M. PERLMAN, *J. Electrochem Soc.* **119**, 892 (1972).
- [14] R. A. CRESWELL, *Dielectric Properties of Solids*, Plenum Press, New York 1972 (p. 295).

(Received March 13, 1985)